

What you can design with this PDK, and where it stops being honest

One capacitance in this technology is measured: $C_{ox} = 1.564 \text{ fF}/\mu\text{m}^2$, from a $400 \times 400 \mu\text{m}$ plate that reads 250.2 pF . The semiconductor's own capacitance - how the channel charges and discharges under the gate - has never been measured here. That single gap decides most of what follows, so this document sorts every kind of design into three boxes and says which box it is in.

Nothing below is quoted from memory. Every number was computed by running ngspice against this PDK's own model files while the document was built; the script that does it is docs/pdf/make_capabilities.py.

	Where it stands	Why
DC design	SOLID	bias, current, swing, gain - all rest on measured DC data
Capacitors	SOLID	C_{ox} is measured and extraction agrees with the drawing to 0.01 %
Gate capacitance	COARSE	C_{ox} times a drawn overlap; no C-V on a transistor exists
Bandwidth, fT	COARSE	good to a factor of two, and the document says which factor
Phase margin, settling	NOT SUPPORTED	they depend on exactly the capacitance nobody measured
Tuned loads	NOT SUPPORTED	the coil's Q is an assumed metal thickness, not a measurement
Matching, offset	NOT SUPPORTED	Kp spans a factor of 70 over 86 devices; no matched-pair data at all

What is measured, and what each measurement licenses

Measured	Value	What it licenses
Cox, MIM plate	$1.564 \text{ fF}/\mu\text{m}^2$	capacitors; overlap capacitance
Kp from ID(VGS,VDS)	$0.717 \text{ uA}/\text{V}^2$ (best)	currents and DC gain
$2 \cdot R_c \cdot W$ from $R_{on}(L)$	$6.6 \text{ MOhm} \cdot \mu\text{m}$	the loss of gm to the contacts
Vto	near 0 V	bias points, with the caveat below
Drawn vs printed	+3 μm , one cell	nothing yet - one sample
NOT measured: gate C-V	-	Cgs, Cgd against bias
NOT measured: matching	-	offset, mismatch, any pair
NOT measured: metal t	-	coil Q, series resistance
NOT measured: stress	-	drift, hysteresis, lifetime
NOT measured: temp.	-	anything over temperature

DC design: what the data actually supports

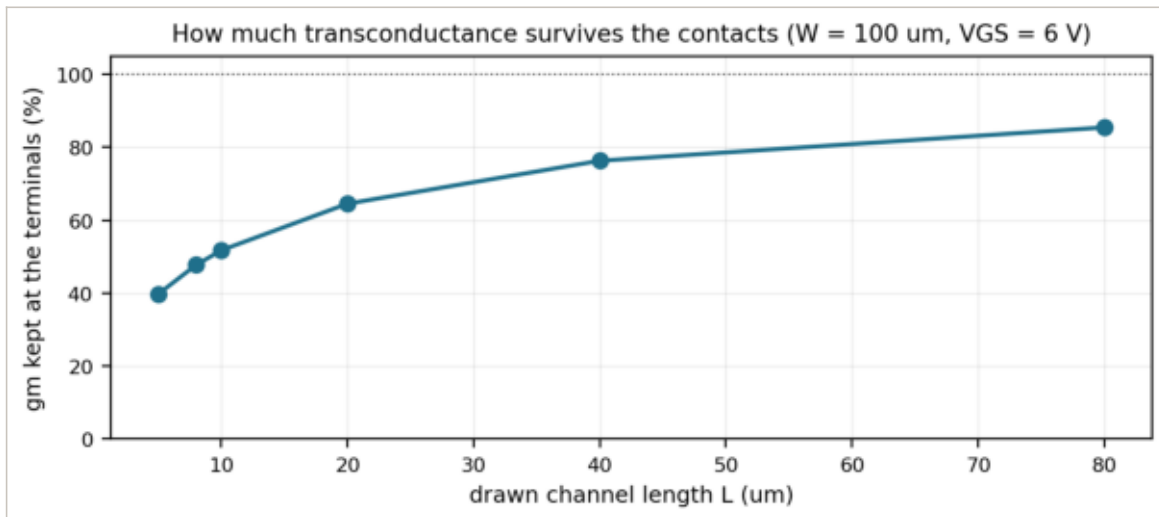
This is the part that works. At $W = 100 \text{ } \mu\text{m}$, $L = 10 \text{ } \mu\text{m}$, $V_{GS} = 6 \text{ V}$, $V_{DS} = 10 \text{ V}$ the tt corner gives $I_D = 57.9 \text{ } \mu\text{A}$, $g_m = 26.6 \text{ } \mu\text{S}$ inside the channel and $g_{ds} = 0.76 \text{ } \mu\text{S}$, so the intrinsic gain g_m/g_{ds} is 35 - about 31 dB for a single stage with an ideal load. Bias points, currents, swing and stage-by-stage DC gain are all reachable from measured data.

Two things to hold on to while doing it. First, the measured saturation exponent is 2.31, not 2, so a square-law model runs about 20 % pessimistic on g_m at a given current - your circuit will do slightly better than it simulates. Second, K_p spans a factor of 70 across the 86 devices measured. Design against a corner, never against a number, and check every result at 'short' and 'all' as well as at 'tt'.

The contacts take half the transconductance, and width will not save you

$2 \cdot R_c \cdot W = 6.6 \text{ M}\Omega \cdot \mu\text{m}$ is a measured number, and the wrapper carries it explicitly as a resistor on each terminal. In saturation the source-side resistor degenerates the stage: at $W = 100 \text{ } \mu\text{m}$, $L = 10 \text{ } \mu\text{m}$ the channel's $26.6 \text{ } \mu\text{S}$ arrives at the terminals as $13.8 \text{ } \mu\text{S}$ - 48 % of it gone.

The trap is thinking wider devices escape it. They do not: $R_c \cdot W$ is a constant, so R_c falls as $1/W$ exactly as g_m rises with W , and the ratio $g_m \cdot R_c$ - the thing that sets the loss - does not depend on W at all. It depends on L . Long channels keep a larger fraction of a smaller g_m ; that is the whole trade.

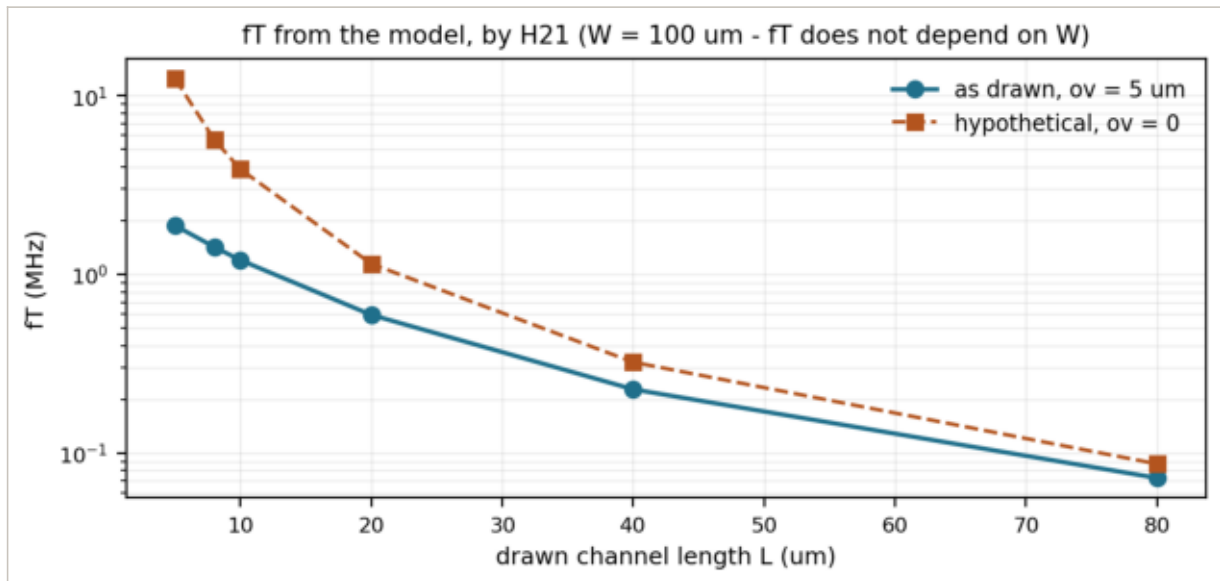


Bandwidth: a real number, with its error bars stated

The gate capacitance is C_{ox} times a drawn overlap plus a textbook channel term. At $W = 100 \text{ } \mu\text{m}$, $ov = 5 \text{ } \mu\text{m}$ that is 0.78 pF of overlap per side - geometry times a measured C_{ox} , which is defensible - and about 1.04 pF of channel, which is the level-1 model's assumption and has never been measured.

Driving the gate with a current and finding where the current gain falls to one gives $f_T = 1.20 \text{ MHz}$ at $L = 10 \text{ } \mu\text{m}$ and 1.88 MHz at $L = 5 \text{ } \mu\text{m}$, the smallest length this process prints. The overlap dominates: with $ov = 0$ the same devices reach 3.9 and 12.5 MHz .

How wrong can this be? Overlap is two thirds of the total capacitance and it is on firm ground. If the channel capacitance were wrong by a factor of two, f_T at $L = 10 \text{ } \mu\text{m}$ would move to about 0.9 MHz or about 2.0 MHz . So the honest statement is 'about a megahertz', and no design should need the second digit.



What not to promise

Phase margin. Settling time. Gain-bandwidth to better than a factor of two. Any compensation scheme whose pole placement depends on Cgs or Cgd. Slew rate into a capacitive load, unless the load is a cap_mim and dwarfs the device. Anything tuned: the inductor's Q rests on an assumed metal thickness, and the same coil is $Q = 2.3$ with a micron of gold and $Q = 0.12$ with 50 nm - the difference between a resonator and a resistor. Matching, offset and any figure that needs two devices to be the same, because no pair has been measured and the population spans a factor of 70.

One C-V sweep on a transistor, gate to source and drain, moves the middle block of this document into the first one. It is the single highest-value measurement left, and it is an afternoon's work on hardware that already exists.

The ceiling, in numbers

Quantity	Value	Set by
Smallest drawn feature	5 μm	the DRC deck, from the fabricated chip
Capacitance per area	1.564 nF/mm ²	measured Cox
Largest capacitor built	250.2 pF	the 400 x 400 μm test plate
Inductance demonstrated	16.4 nH	8-turn square coil, DRC clean
Coil Q at 100 MHz	2.3 / 0.12	1 μm of gold / 50 nm of gold
Device fT	1.2 - 1.9 MHz	overlap capacitance, then L
Model validity	VGS $\leq$ 6 V, VDS $\leq$ 10 V	the measured range
Temperature	300 K only	nothing was measured anywhere else

Two changes to the model this round

Source and drain are now interchangeable

The wrapper used to tie the intrinsic bulk to the source pin, so a device drawn with its source on the high side forward-biased the bulk-drain junction and conducted through it: 654 uA one way round, 1414 uA the other. A real TFT has no body and its two channel terminals are the same kind of terminal. Every junction on all four model cards is now inert - IS, JS, CBD, CBS, CJ and CJSW all zero - which leaves the intrinsic substrate node with no path anywhere, and the wrapper's own parts were already symmetric. Swapping d and s now changes nothing, and the check script asserts it at every corner, in saturation and in the linear region.

Existing schematics keep working and keep their numbers. What changes is that orientation stops being something a designer has to remember.

A magnetic field, defaulting to zero

igzo_tft now takes B in tesla, default 0, and a dimensionless b_scale, default 1. The channel's mobility falls as $\mu/(1 + (\mu*b_scale*B)^2)$ - classical magnetoresistance - with μ taken from the active corner's own K_p/Cox , so the corners respond differently, as they should.

Its size, at the measured $\mu = 4.58 \text{ cm}^2/\text{Vs}$: $\mu*B = 4.6\text{e-}4$ per tesla, so the effect on the current is $(\mu*B)^2 = 2$ parts in ten million per tesla. That is not a sensor. It is not measurable. If the lab measures a reduction in K_p under a field, classical magnetoresistance is not the cause, and docs/bibliography/ says what is published, what it would predict here, and which ordinary artefacts - self-heating, stress, bias hysteresis, the magnet acting on the instrument - to rule out first. Reverse the field: a real mobility effect is even in B.

When a coefficient has been measured, set $b_scale = \sqrt{\text{measured} / 2.1\text{e-}7}$ and the model reproduces it without anyone editing a corner file. A measured 1 % per tesla puts b_scale near 220 - which is also the factor by which such a measurement would exceed the physics the term is named after.